SQL Aggregate Functions and Operators

1) The students table

1) SELECT DISTINCT department
FROM students

department
IT
HR
Finance

2) Averge age per department

SELECT department, AVG(age) AS avg-age
FROM students
GROUP BY department;

department	avg-age
IT	21.5
HR	21
Finance	22

3) Departments with more than 1 student

```
SELECT department, AVG(age) AS avg_age, COUNT(*) AS Student_count
FROM students
GROUP BY department
HAVING COUNT(*)>1;
```

Department	avg_age	Student_count
IT	21.5	2

4) Students aged between 21 and 23

```
SELECT student_id, name, age
FROM students
WHERE age BETWEEN 21 AND 23;
```

student_id	name	age
2	Bob	22
3	Charlie	21
4	Diana	23
5	Eve	22

5) IT or HR students ~~the~~ older than 21

```
SELECT student_id, name, age, department
FROM students
WHERE department IN ('IT', 'HR')
AND age > 21;
```

student_id	name	age	department
2	Bob	22	IT
4	Diana	23	IT

6) Total credits per department

```
SELECT department, SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING sum(credits) > 5;
```

↓

department	total_credits
IT	11

7) Courses not equal to 4 credits

```
SELECT course_id
FROM courses
WHERE credits <> 4;
```

↓

course_id
101
102
105

8) Top 3 courses by credits

```
SELECT course_id, course_name, department, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;
```

course_id	course_name	department	credits
103	Data Science	IT	4
104	Excel	Finance	4
101	SQL Basics	IT	3

9) Max, Min, Avg grade

```
SELECT
MAX (grade) AS max_grade;
MIN (grade) AS min_grade
AVG (grade) AS avg_grade
FROM enrollments;
```

max_grade	min_grade	avg_grade
90	78	84.6

10) Enrollments per course

```
SELECT course_id, COUNT(*) AS enrollment_count
FROM enrollments
GROUP BY course_id;
```

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

11) Total Salary & bonus per department

```
SELECT department,
Sum (salary) AS total_salary
Sum (bonus) AS total_bonus
FROM salaries
GROUP BY department;
```

department	total_salary	total_bonus
IT	122000	11 000
HR	109000	8 500
Finance	70 000	6000

12) Departments with avg salary > 55,000

```
SELEC department, AVG (salary) AS avg_salary
FROM salaries
Group BY department
HAVING AVG (salary) > 55 000;
```

department	avg_salary
IT	61 000
Finance	70 000

13) Employees with total comp > 60,000

```
SELECT employee_id, name, salary, bonus, (salary + bonus) AS
       total_compensation
FROM salaries
WHERE (salary + bonus) > 60 000;
```

employee_id	name	salary	bonus	total_compensation
1	Tom	60 000	5000	65 000
3	Spike	70 000	4000	74 000
4	Tyke	62 000	6 000	68 000

14) Total and avg budget per department (> 70,000 avg)

```
SELECT department,
Sum(budget) AS total_budget,
AVG (budget) AS avg_budget
FROM projects
GROUP BY departments
HAVING AVG(budget) > 70 000;
```

department	total_budget	Avg_budget
IT	200 000	100 000
Marketing	150 000	150 000

15) Projects between 50k - 120k (not Marketing)

```
SELECT project_name, department, budget
FROM projects
WHERE budget BETWEEN 50000 AND 120000
AND department <> 'Marketing';
```

↓

PROJECT_NAME	DEPARTMENT	BUDGET
AI APP	IT	120000
Payroll System	Finance	80000
HR Portal	HR	6000